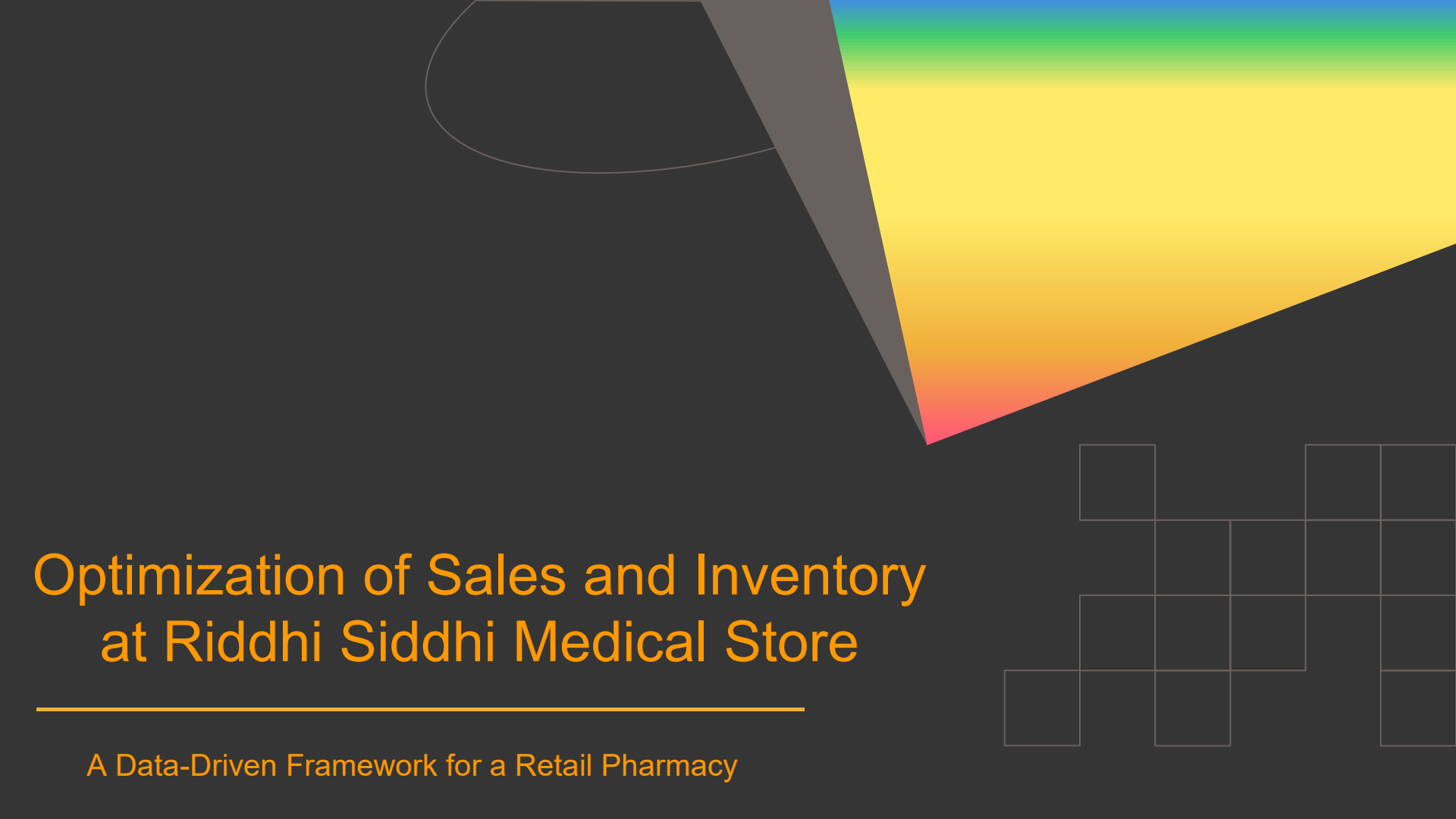




Optimization of Sales and Inventory at Riddhi Siddhi Medical Store

A Data-Driven Framework for a Retail Pharmacy

Riddhi Siddhi Medical Store

- **Location:** Kasba Bansur, Alwar, Rajasthan
- **Venue:** Retail Pharmacy
- **Owner:** Actively managed by the owner
- **Products:** Branded and generic medicines, prescription and over-the-counter (OTC) drugs, and healthcare supplements

Key Features:

- **Established:** December 2021
- **Market Environment:** Operates in a highly competitive local market with over 25 medical shops and 10 hospitals in the vicinity
- **Financial Situation:** The business faces declining sales and financial losses due to inefficient inventory management and stock expiry



Objective:

- Utilize advanced data analysis to optimize sales and inventory management at Riddhi Siddhi Medical Store.

Project Focus:

- **Tackling Declining Sales:** Enhance revenue through predictive sales forecasting (XGBoost) and by identifying the most profitable and efficient medicines using DEA and RFM analysis.
- **Optimizing Inventory Management:** Minimize financial losses by analyzing overstocking, understocking, and expiry waste, and by implementing value-based controls through ABC analysis.

Data Source:

Primary data was sourced from the Riddhi Siddhi Medical Store's internal sales and inventory records from **November 2024 to January 2025**, and compiled into `sales_data.csv` and `inventory_data.csv`.

Data Preparation:

The raw data was prepared for analysis by cleaning inconsistencies (like in 'Medicine Name'), converting date columns to a proper datetime format, and removing statistical outliers using the Interquartile Range (IQR) method.

Data-Driven Focus

Uncovering Deep Inventory Inefficiencies

Key Findings:

- **Significant Wastage:** 16.4% of the pharmacy's expired stock was never sold, resulting in a direct financial loss.
- **Capital Mismatch:** There was a major imbalance between stock levels and actual sales. Items like **Metol XI-25** were heavily overstocked, while high-demand products like **Paracetamol** were understocked.

Insights:

- **Financial Drain:** The unsold expired inventory points to a tangible and preventable financial drain caused by poor stock rotation or inaccurate forecasting.
- **Inefficient Capital:** Overstocking ties up capital in slow-moving products and increases expiry risk, while understocking leads directly to lost sales and customer dissatisfaction.

Implications:

- **Improve Stock Rotation:** Implement a strict **First-Expiry-First-Out (FEFO)** policy to minimize wastage from expired products.
- **Data-Driven Replenishment:** Fine-tune inventory and reorder policies based on real sales trends to prevent both stockouts and excess inventory.

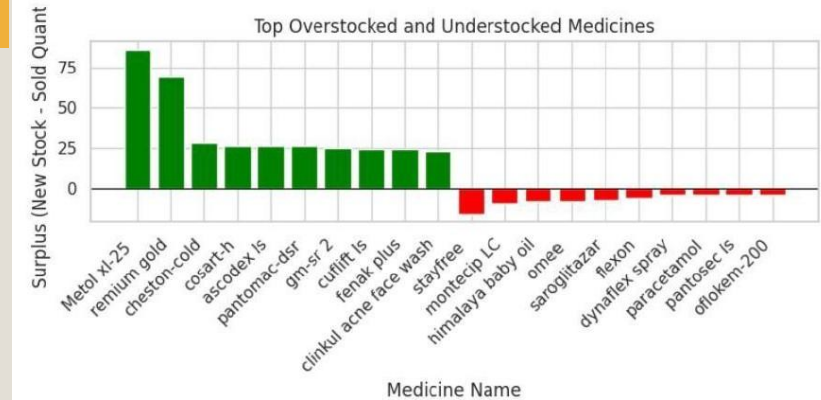


Figure 1: Top Overstocked and Understocked Medicines

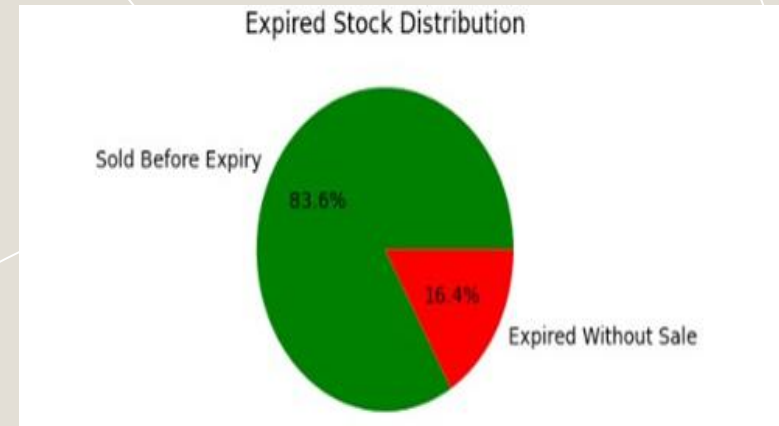


Figure 2: Expired Stock Distribution

Finding 2: Identifying High-Value Products (ABC & RFM)

Key Findings:

- **ABC Analysis:** The bar chart on the top visually confirms the **Pareto principle**; a few high-value **"A" items** (in red) account for the majority of the total inventory value.
- **RFM Analysis:** The bubble chart on the bottom segments products by performance, clearly identifying the **"sales drivers"** (large bubbles, bottom-right) from the **"stagnant stock"** (small bubbles, top-left).

Insights:

- The concentration of value in a few "A" items proves that a **granular, product-level focus** is more effective than managing by broad categories.
- A one-size-fits-all inventory strategy is inefficient because the RFM chart shows that a small subset of medicines generates the most consistent revenue.

Implications:

- Implement **product-level controls** to manage the true high-value "A" items closely, ensuring they are always prioritized.
- Adopt a **tiered inventory strategy**: ensure "sales drivers" are always in stock while actively clearing "stagnant stock" with discounts or discontinuation.

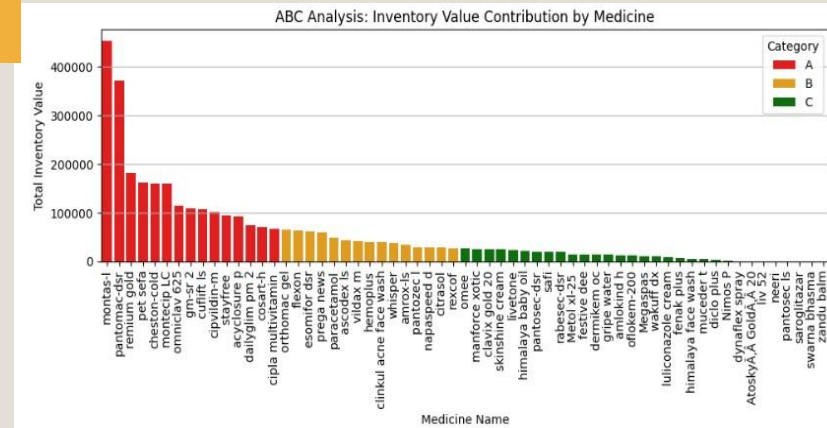


Figure 3:ABC Analysis Inventory Value Contribution by Medicine

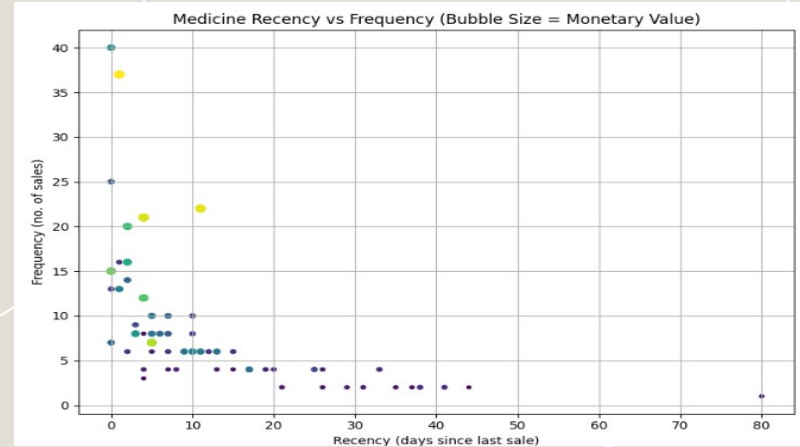


Figure 4: Medicine Recency vs Frequency (Bubble Size = Monetary Value)

Finding 3: Measuring True Product Profitability (DEA)

Key Findings:

- **Widespread Inefficiency:** The majority of medicines have low DEA efficiency scores, with most falling between **0.0 and 0.4**.
- **Profitability Champions:** A small number of products, like **Saroglitzar** and **Swarna Bhasma**, are highly efficient, with scores near **1.0**, indicating ideal performance.

Insights:

- Most of the inventory investment is not being converted into profit effectively, pointing to widespread inefficiency in the product mix.
- The store's overall profitability is heavily dependent on a few key "efficiency champion" products that provide the best possible financial return.

Implications:

- Re-evaluate low-efficiency products by reducing their stock, renegotiating purchase prices, or using promotions to improve their turnover.
- Prioritize and protect the investment in "efficiency champion" products to ensure they are always in stock, which will maximize overall profit.

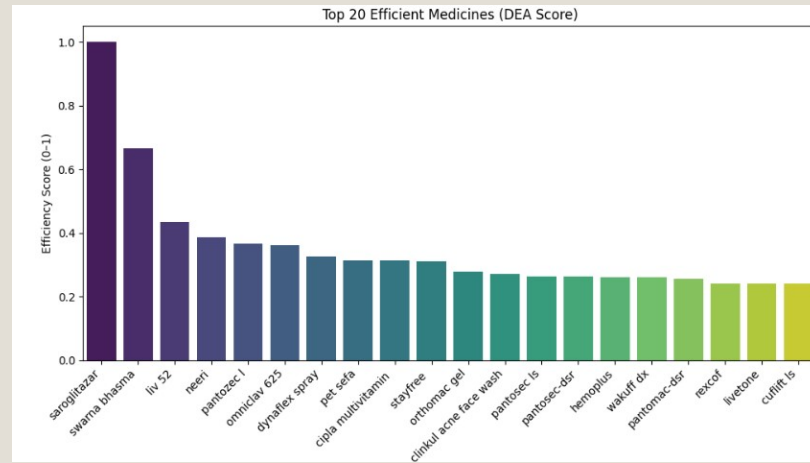


Figure 5: Top 20 Efficient Medicines (DEA Score)

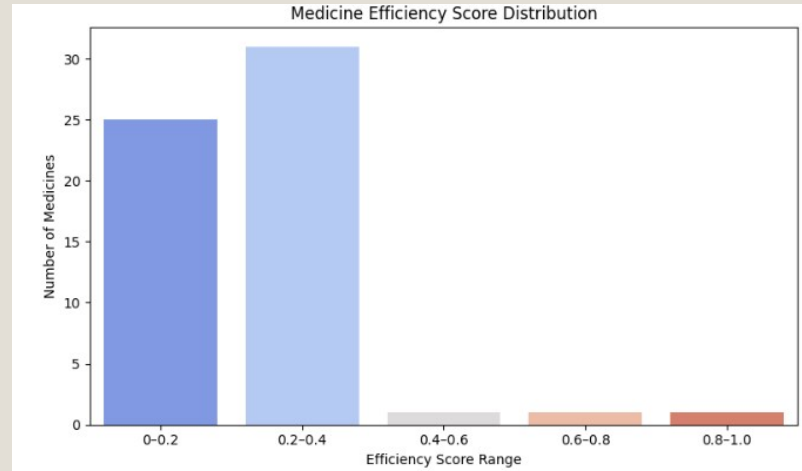


Figure 6: Medicine Efficiency Score Distribution

Finding 4: Achieving Accurate Sales Forecasting (XGBoost)

Key Findings:

- **High Accuracy:** The final XGBoost model successfully predicted daily sales trends with a strong **R² score of 0.7868**.
- **Model Reliability:** The analysis of residuals shows a near-normal distribution centered around zero, confirming that the model's predictions are stable and not systematically biased.

Insights:

- The model can explain approximately **79% of the variance** in daily sales, making it a powerful and reliable tool for the business.
- The lack of bias in the model's errors means the business can trust its forecasts for strategic planning without worrying about consistent over or under-prediction.

Implications:

- The business should deploy this model to generate daily or weekly sales forecasts, shifting from reactive to proactive decision-making.
- These forecasts should be directly integrated into inventory and marketing planning to prevent stockouts of popular items and optimize promotional timing.

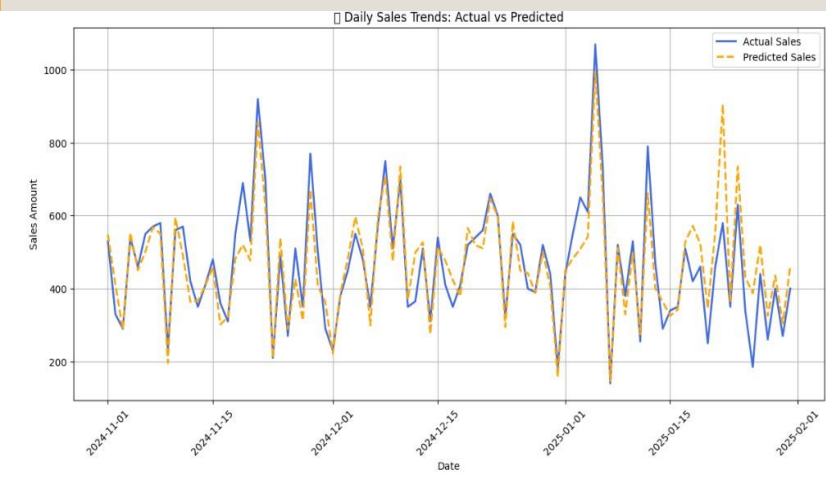


Figure 7: Daily Sales Trends: Actual vs Predicted

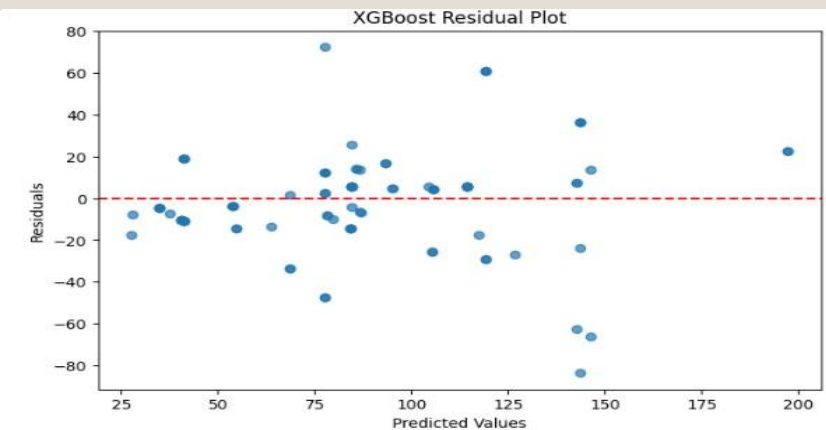


Figure 8:XGBoost Residual Plot

Recommendations

Predictive Sales Forecasting

- Deploy the XGBoost model ($R^2 \approx 0.79$) to generate daily and weekly sales forecasts to guide planning.
- Integrate these forecasts into procurement cycles to prevent stockouts of high-demand items and optimize promotional timing.

Tiered Inventory & Promotion Strategy

- Use ABC and RFM analysis to identify high-value "sales drivers" and low-value "stagnant stock".
- Prioritize keeping "sales drivers" consistently in stock and use targeted discounts to clear "stagnant stock", freeing up capital.

Waste Reduction & Stock Balancing

- Implement a strict FEFO (First-Expiry-First-Out) system with automated alerts to reduce the 17% expiry loss .
- Use dynamic dashboards to monitor the surplus/deficit of key medicines in real-time, preventing both overstocking and understocking.

Profit-Driven Procurement

- Use DEA efficiency scores as a core metric during purchase planning to identify the most profitable products .
- Prioritize investment in high-efficiency "profitability champions" and re-evaluate or reduce stock of underperforming items

Conclusion: Insights and Strategic Actions

Key Insights from Analysis:

- **Quantified Inefficiency:** The analysis quantified significant financial drain from a **17% expiry loss** and a major mismatch between stock levels and actual sales.
- **Identified Key Products:** It pinpointed the most valuable (ABC), best-selling (RFM), and most profitable (DEA) products, proving that value is concentrated in a small subset of the inventory.
- **Achieved Predictive Accuracy:** A reliable **XGBoost sales forecasting model** was successfully developed, achieving a strong **R² score of 0.7868**.

Strategic Actions for Store Enhancement:

- **Data-Driven Forecasting:** Deploy the XGBoost model to shift from reactive to proactive demand and inventory planning.
- **Performance-Based Inventory:** Implement a tiered strategy based on ABC, RFM, and DEA to focus capital on high-performing products.
- **Waste Elimination:** Implement strict **FEFO** logic and dynamic dashboards to reduce losses from expiry and stock imbalances.

Moving Forward:

- Implement these insights to reduce operational costs, enhance revenue, and improve overall profitability.
- Leverage this data-driven framework to gain a sustainable competitive advantage in the local market.



Thank you